



MOIZ DAWOODI

O/A LEVEL PHYSICS

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The Nuclear Model of an Atom

5.1

STUDENT NAME



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PHYSICS WITH
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SYLLABUS OUTLINE:

5.1 The nuclear model of the atom

5.1.1 The atom

- 1 Describe the structure of the atom in terms of a positively charged nucleus and negatively charged electrons in orbit around the nucleus
- 2 Describe how alpha-particle scattering experiments provide evidence for:
 - (a) a very small nucleus surrounded by mostly empty space
 - (b) a nucleus containing most of the mass of the atom
 - (c) a nucleus that is positively charged

5.1.2 The nucleus

- 1 Describe the composition of the nucleus in terms of protons and neutrons
- 2 Describe how atoms form positive ions by losing electrons or negative ions by gaining electrons
- 3 Define the terms proton number (atomic number) Z and nucleon number (mass number) A and be able to calculate the number of neutrons in a nucleus
- 4 Explain the term nuclide and use the nuclide notation A_ZX
- 5 Explain what is meant by an isotope and state that an element may have more than one isotope

5.1.1

The Atom

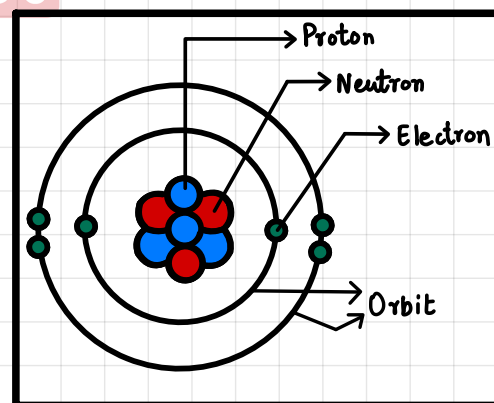
Atom.

"Atom is the smallest particle of an element that can exist."

The center of an atom is known as nucleus.

There are three sub atomic particles that make an atom.

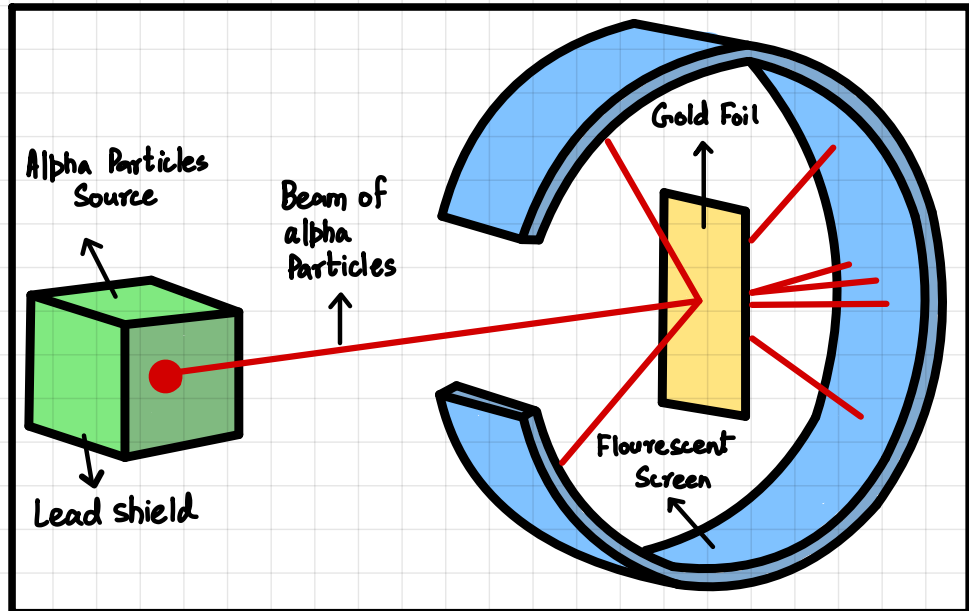
- **Electron:** Negatively charged particle that resides in orbits around a nucleus.
- **Proton:** Positively charged particle that resides inside a nucleus.
- **Neutron:** Neutral particle that resides inside a nucleus.



GEIGER-MARSDEN ALPHA-SCATTERING EXPERIMENT:

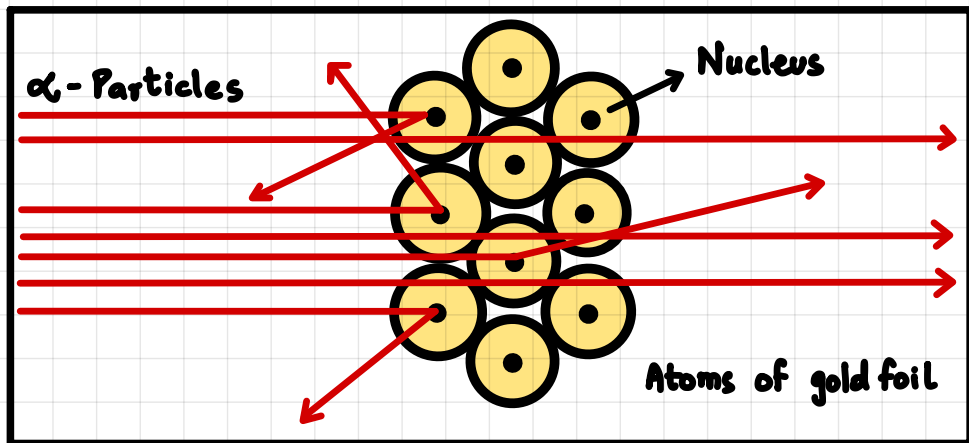
OBSERVATION:

Alpha particles are allowed to pass through thin gold foil. Alpha particles get scattered at different angles & produce fluorescence on fluorescent screen.



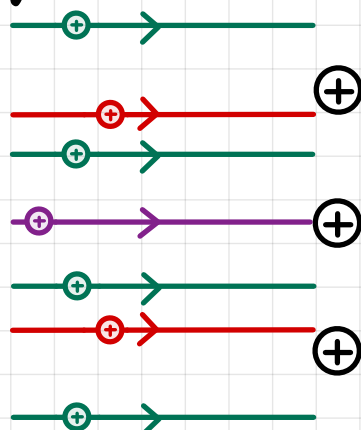
OUTCOMES:

- α -Particles bending occurs which proves that nucleus is positively charged.
- The size of the nucleus is very small as compared to atom as most of the alpha particles went undeflected.
- Atom is highly vacant & most of the mass of atom is inside nucleus.



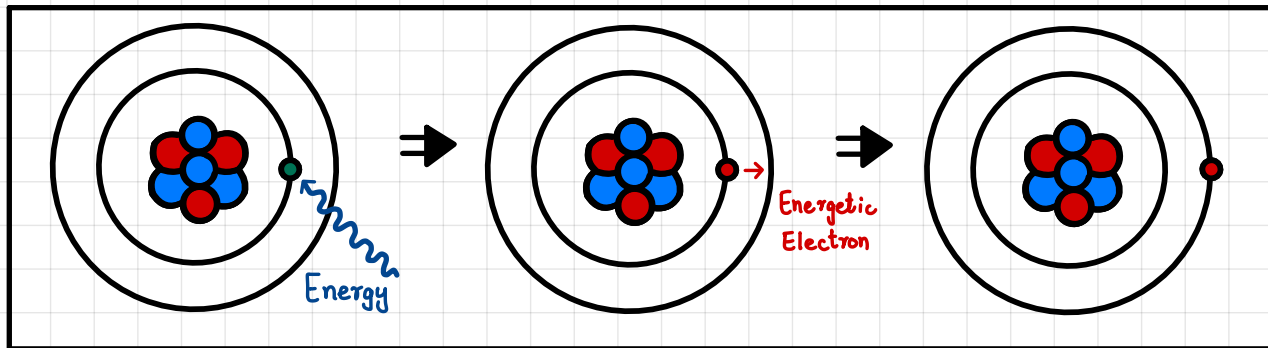
EXERCISE.

Draw the path of alpha particles as they go near the nucleus.

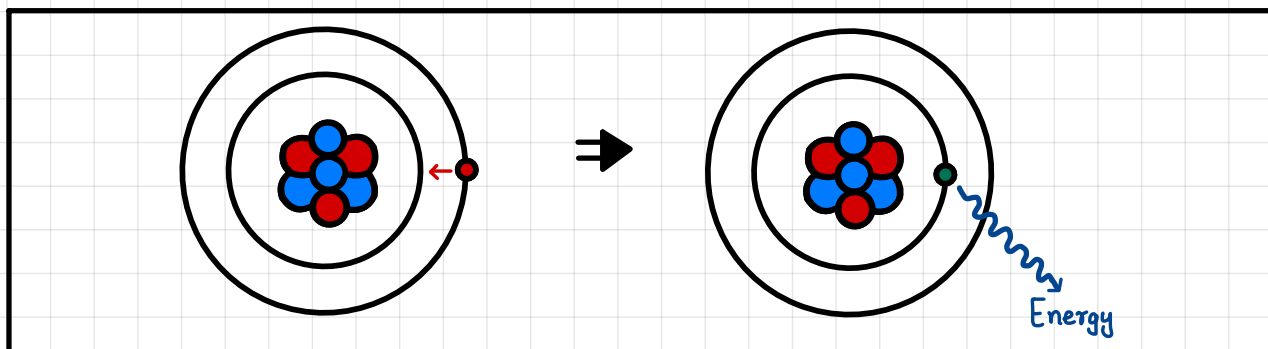


IONIZATION ENERGY:

Normally electrons reside in their ground orbits but when energy is provided to electrons, they receive it & jump towards outer orbits.



Soon afterwards electron returns to its ground orbit & release energy in the form of electromagnetic radiations called photons.



If an atom gains enough energy for an electron to move out of an atom, the atom becomes an ion & the energy needed to do this is called ionization energy.

"Energy required to eject electron out of the atom is called ionization energy."

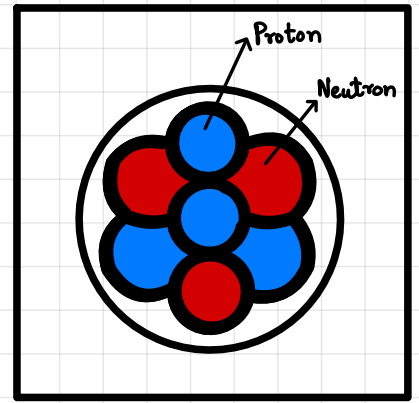
ION.

"When an electron is lost, atom becomes positive ion & when an electron is gained, atom becomes negative ion."

NUCLEUS:

"Nucleus is very small, dense region at the center of an atom containing proton & neutron."

- Both proton & neutron reside inside the nucleus & are called nucleons.



PROTON NUMBER:

"Number of protons present inside the nucleus is called Proton Number."

It is also called Atomic Number.

It is denoted by 'Z'.

Mass of Proton $\cong$ Mass of Neutron

NUCLEON NUMBER:

"Sum of proton & neutrons inside the nucleus is called Nucleon Number."

It is also called Mass Number.

It is denoted by 'A'.

Number of Neutrons

$$N = A - Z$$

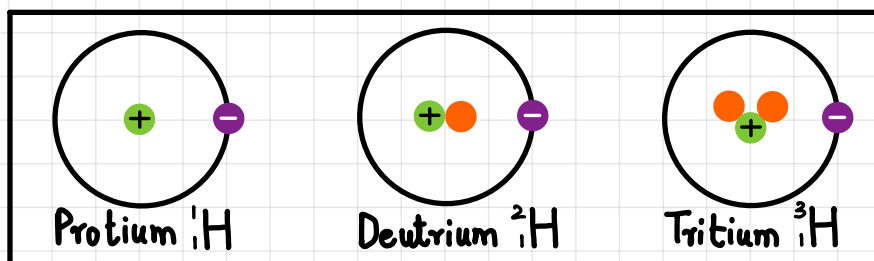
NUCLIDE NOTATION:

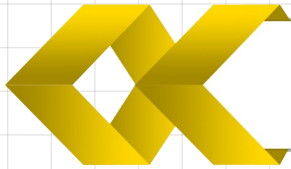
"Nuclide notation is a short hand method of showing information about atom."



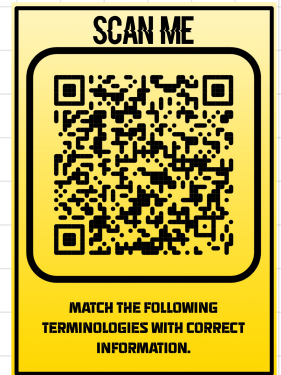
ISOTOPE:

"Atoms of same elements having same proton numbers but different neutron numbers are called isotopes."





Match the following



Electron is

neutral particle.

Proton is

positively charged particle.

Neutron is

negatively charged particle.

Carbon $^{14}_6\text{C}$ has

same proton number but different nucleon number.

Nucleus is highly

empty & smaller than atom.

Isotopes have

6 protons & 8 neutrons.

Electron resides in

reside inside nucleus.

Neutron & Proton

orbits around nucleus.